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# **Module 3 Lecture - Biopsychology**

Introductory Psychology

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# 1 Overview and Introduction

## 1.1 Textbook Learning Objectives

- Explain how scientific research addresses questions about behavior
- Discuss how scientific research guides public policy
- Appreciate how scientific research can be important in making personal decisions
- Describe the different research methods used by psychologists
- Discuss the strengths and weaknesses of case studies, naturalistic observation, surveys, and archival research
- Compare longitudinal and cross-sectional approaches to research
- Compare and contrast correlation and causation
- Explain what a correlation coefficient tells us about the relationship between variables
- Recognize that correlation does not indicate a cause-and-effect relationship between variables
- Discuss our tendency to look for relationships between variables that do not really exist
- Explain random sampling and assignment of participants into experimental and control groups
- Discuss how experimenter or participant bias could affect the results of an experiment
- Identify independent and dependent variables
- Discuss how research involving human subjects is regulated
- Summarize the processes of informed consent and debriefing
- Explain how research involving animal subjects is regulated

## 1.2 Instructor Learning Objectives

- Understand the purpose on learning anatomy to contextualize human behavior and cognition
- Appreciate the complexity in our nervous system processes and how they interact to rise to cognition

## 1.3 Introduction

- Brains and nerves are strange things, but they are the                      basis for our thoughts and behaviors - so it behooves us to study and >understand them
- This module will help introduce the many physiological and                                      processes that are associated with what we do, think, and feel

 Discuss: There is a broader philosophical debate on whether humans are purely biological and chemical beings, i.e., do we have a soul? or are we just being driven by chemical equations and processes - what do you think?

## 2 Human Genetics

### 2.1 Introduction

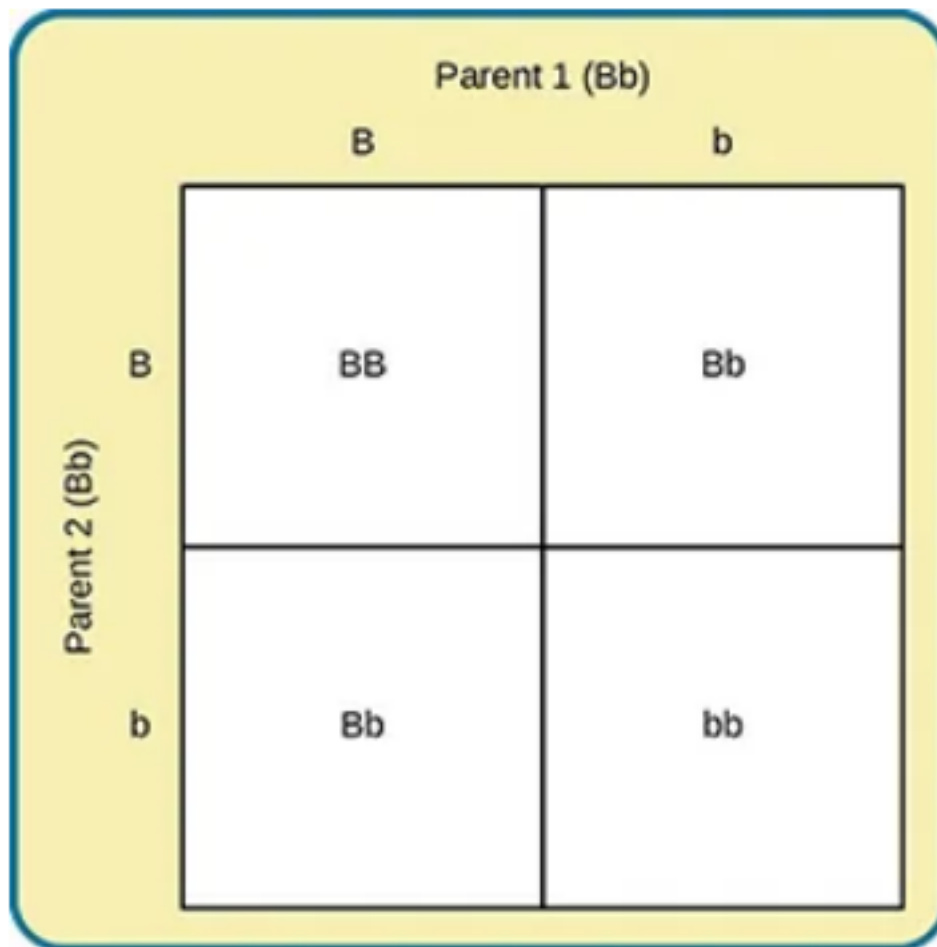
- Our genetics are the traits we \_\_\_\_\_ from our biological parents, that play into the specific and unique code that make up the basis of cells in our body
  - In this case, I'm using "parents" to refer to those who contribute the \_\_\_\_\_, i.e., sperm and egg, that create the zygote
- Understandably, genetics play a very \_\_\_\_\_ role in our physical health and characteristics, but also are associated with our psychological make-up as well
  - A basic \_\_\_\_\_ example: height - if you have two tall parents, you are likely to be tall, if you have two shorter parents, you are likely to be short
  - A \_\_\_\_\_ example: If both of your parents have bipolar disorder, are you more likely to? It's complicated...

### 2.2 Genetic Variation

- Variation in our genetics contributes to the \_\_\_\_\_ differences that we notice between each person
  - Our hair, skin, and eye color
  - The dimensions and structure of our face
  - \_\_\_\_\_ diseases or conditions

 Discuss: What are some other examples of physical/mental traits that are genetic, i.e., passed down by parents?

- The specific \_\_\_\_\_ that each person has is a function of the sperm and egg cells that come together at the beginning of pregnancy
  - Both the \_\_\_\_\_ and sperm cells have 23 **chromosomes** made up of **DNA**, the building blocks of our being
  - The sperm and egg cells combine for a total of 46 \_\_\_\_\_ into a zygote, which has a specified **genotype**, or genetic make up. That genotype contributes to the **phenotype**, or the apparent physical characteristics that manifest.
  - Specific \_\_\_\_\_ and sections of that DNA make up **genes**, which function sort of like switches that produce certain outcomes. These genes can be in different variations, or **alleles**, thus resulting in different effects
- Most physical traits are complex and result from \_\_\_\_\_ different genes, which would be called **polygenic** traits
  - However some genes are slightly \_\_\_\_\_, such as having a cleft chin, in which there is only one gene that contributes.
  - Having two of the \_\_\_\_\_ allele from both parents is called **homozygous**, whereas having two \_\_\_\_\_ alleles is referred to as being **heterozygous**.
  - A **dominant** allele is one that “overrides” a **recessive** allele to become the phenotype; a \_\_\_\_\_ allele will only reflect in a phenotype if both alleles are recessive.



- **Mutation** comes from a sudden and change to the genes that result in changes to the genotype - which could manifest as disadvantageous, deadly, or useful phenotypes

? Mutation can lead to increased 'fitness' or ability to adapt / survive. Which psychological perspective would focus intensely on this, outside of biopsychology?

- ☒ A) Cognitive
- ☐ B) Behavioral
- ☐ C) Gestalt
- ☐ D) Evolutionary

Explanation:

## 2.3 Gene-Environment Interactions

- Genes are not \_\_\_\_\_ deterministic - while they do have a meaningful impact, they only explain one part of our being

### ! Important

There are several different perspective on how genes and the environment interact with one another - carefully consider the following as different perspectives, and think about which you may align with

- Genes can understood as setting a **range of reaction**, or effectively, a range of \_\_\_\_\_ outcomes based on our environment. This means that genes, in a way, express what our potential is.
  - It is believed that our \_\_\_\_\_ plays into how much of that potential is realized or not
- Others may posit that genes and the environment are \_\_\_\_\_, so that our genes drive us towards a certain environment, which in turn reinforces a certain expression of our genes
  - This sort of \_\_\_\_\_ relationship means that both our genes and environment play an equal role in our outcomes
- Yet another perspective comes from the field of **epigenetics**, which is concerned with how the \_\_\_\_\_ genotype can give rise to different phenotypes, as a result of our environment
  - Example: instructor's identical twin
- Many students are introduced to genes by studying and practicing on physical characteristics, largely because they are relatively \_\_\_\_\_ to identify and measure
  - However, genes also play a role in the mental, like our intellect, mental health, personality, etc.
  - But, given the \_\_\_\_\_ of mental characteristics, the direct impact of genes can be much harder to parse out
- This section is connected to the concept of **nature vs. nurture**, something we will revisit during the \_\_\_\_\_ psychology module

## 3 Cells of the Nervous System

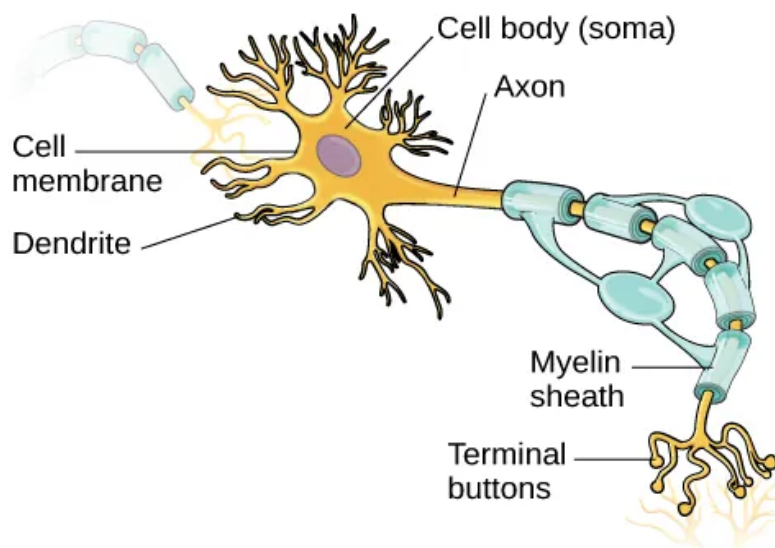
### 3.1 Introduction

- The human nervous system is comprised of two types of specialized cells:
  - The **neuron**: which is the \_\_\_\_\_ communication unit which conveys signals to other neurons and organs

- The **glial cell**: which supports neurons cell functions
- There is roughly an equal amount of these two types of cells through the nervous system in humans
  - Give or take, around 100 \_\_\_\_\_ neurons in an adult brain

### 3.2 Neuron Structure

- A neuron is a \_\_\_\_\_ cell that has a **semi-permeable membrane** allowing certain molecules to pass through its outer surface to create a charge



- The elongated **axons** are covered in a **myelin sheath**, separated by **Nodes or Ranvier**. The myelin sheath serves a purpose to \_\_\_\_\_ and preserve the special electric signal traveling through the axon.
  - \_\_\_\_\_ such as Multiple sclerosis are called “demyelinating diseases” because they degrade this insulation, which has compounding negative physical impacts
- The **terminal buttons** at the end of each neuro contains **synaptic vessels**, which in turn, have **neurotransmitters** that serve as a method of communication with other neurons’ \_\_\_\_\_.
  - When a charge \_\_\_\_\_ down the length of the axon, it triggers actions in the terminal buttons
- When triggered, neurotransmitters are released into the **synaptic cleft**, between the synaptic vessels of the transmitting neuron, and the dendrites with **receptors** on the receiving neuron(s)
  - Specific neurotransmitter bind to \_\_\_\_\_ sites on the receptors, like a key going into a keyhole

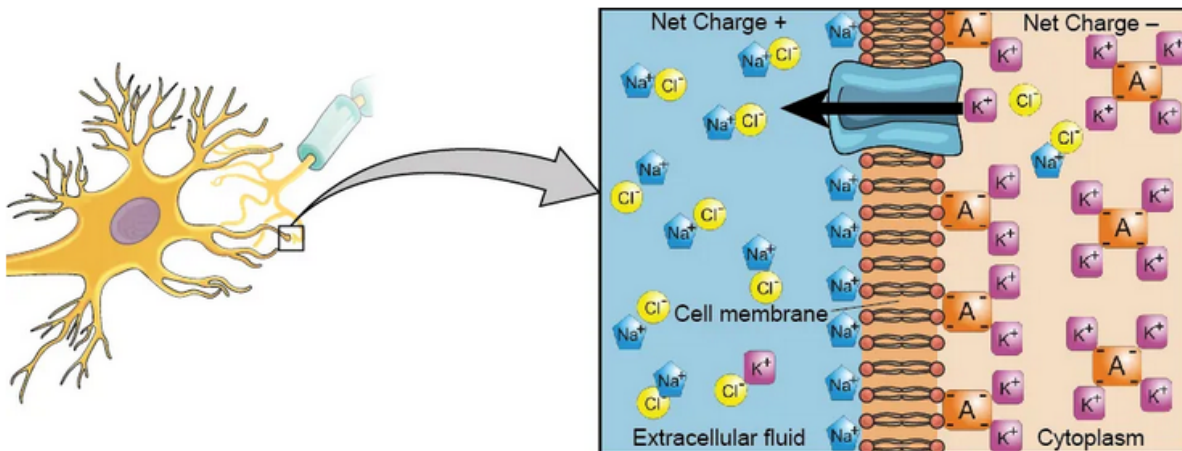


### 3.3 Neuronal Communication

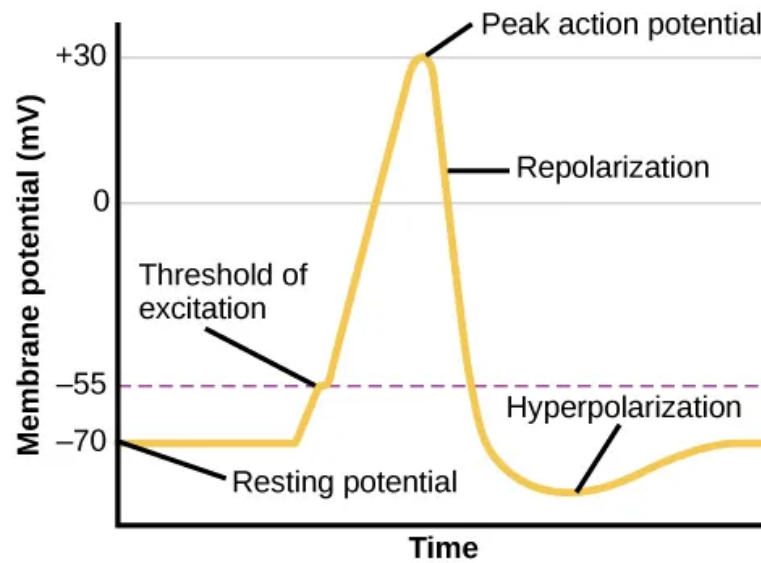
#### ! Important

Get ready for some light chemistry y'all - from your chemistry-incompetent instructor

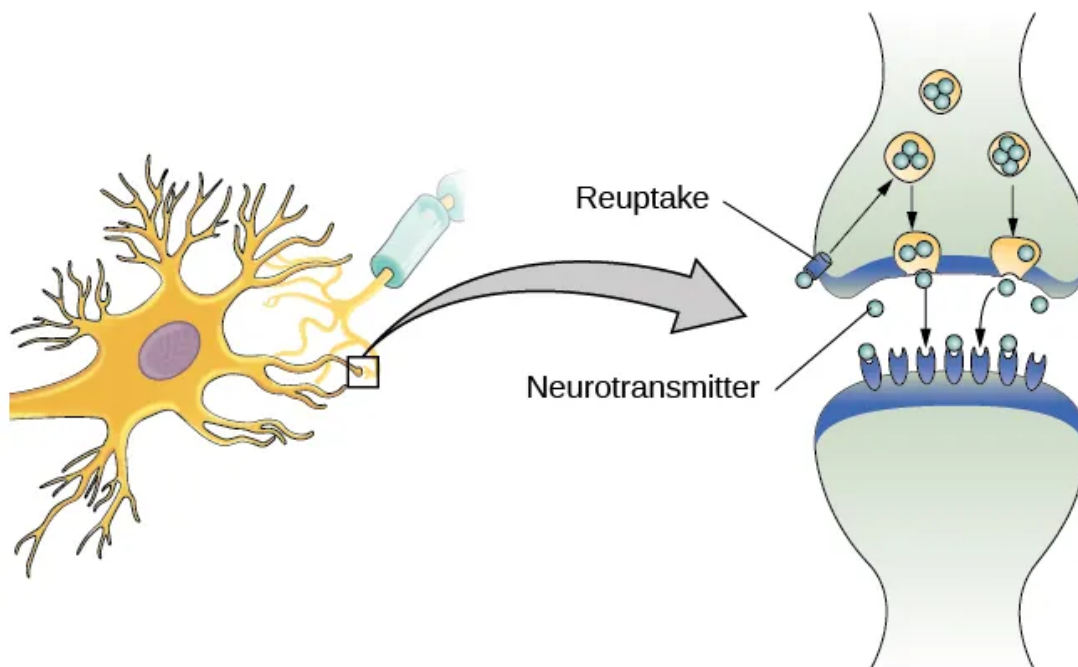
- A signal from the neuron is created by a **membrane potential** resulting from a change in concentration of charged particles across the semi-permeable membrane
- At a baseline, the neuron holds a **resting potential** where there is an imbalance of \_\_\_\_\_ particles on one side of the membrane, so that the neuron remains “ready” to fire
  - The primary particles for us to pay attention to are the positively-charged sodium ions ( $\text{Na}^+$ ) and the positively-charged potassium ions ( $\text{K}^+$ ), and their relative concentrations inside and outside the cell



- The neuron receives a \_\_\_\_\_ via its dendrites, and quickly undergoes rapid change to allow an influx of  $\text{Na}^+$ , greatly increasing the positive energy of the cell - this is an **all-or-nothing** process



- The action potential reaches the terminal buttons, triggering neurotransmitters to push into the synaptic cleft, where they are received by other neurons, which in turn push their own  potential
  - Unused particles in the synaptic cleft return to the original neuron, during **reuptake**



### 3.4 Neurotransmitters and Drugs

- There are many types of neurotransmitters, which serve \_\_\_\_\_ purposes and are launched for specific types of signals

#### Major Neurotransmitters and How They Affect Behavior

| Neurotransmitter               | Involved in                  | Potential Effect on Behavior            |
|--------------------------------|------------------------------|-----------------------------------------|
| Acetylcholine                  | Muscle action, memory        | Increased arousal, enhanced cognition   |
| Beta-endorphin                 | Pain, pleasure               | Decreased anxiety, decreased tension    |
| Dopamine                       | Mood, sleep, learning        | Increased pleasure, suppressed appetite |
| Gamma-aminobutyric acid (GABA) | Brain function, sleep        | Decreased anxiety, decreased tension    |
| Glutamate                      | Memory, learning             | Increased learning, enhanced memory     |
| Norepinephrine                 | Heart, intestines, alertness | Increased arousal, suppressed appetite  |
| Serotonin                      | Mood, sleep                  | Modulated mood, suppressed appetite     |

- Certain medication work to modify behavior in the synaptic cleft:
  - Agonist** medications \_\_\_\_\_ and recreate neurotransmitters so that neurons receive greater signals
  - Antagonists** \_\_\_\_\_ binding of neurotransmitters to receptors to prevent over-consumption
  - Reuptake inhibitors** \_\_\_\_\_ reuptake, to help certain neurotransmitter remain longer in the cleft

## 4 Parts of the Nervous System

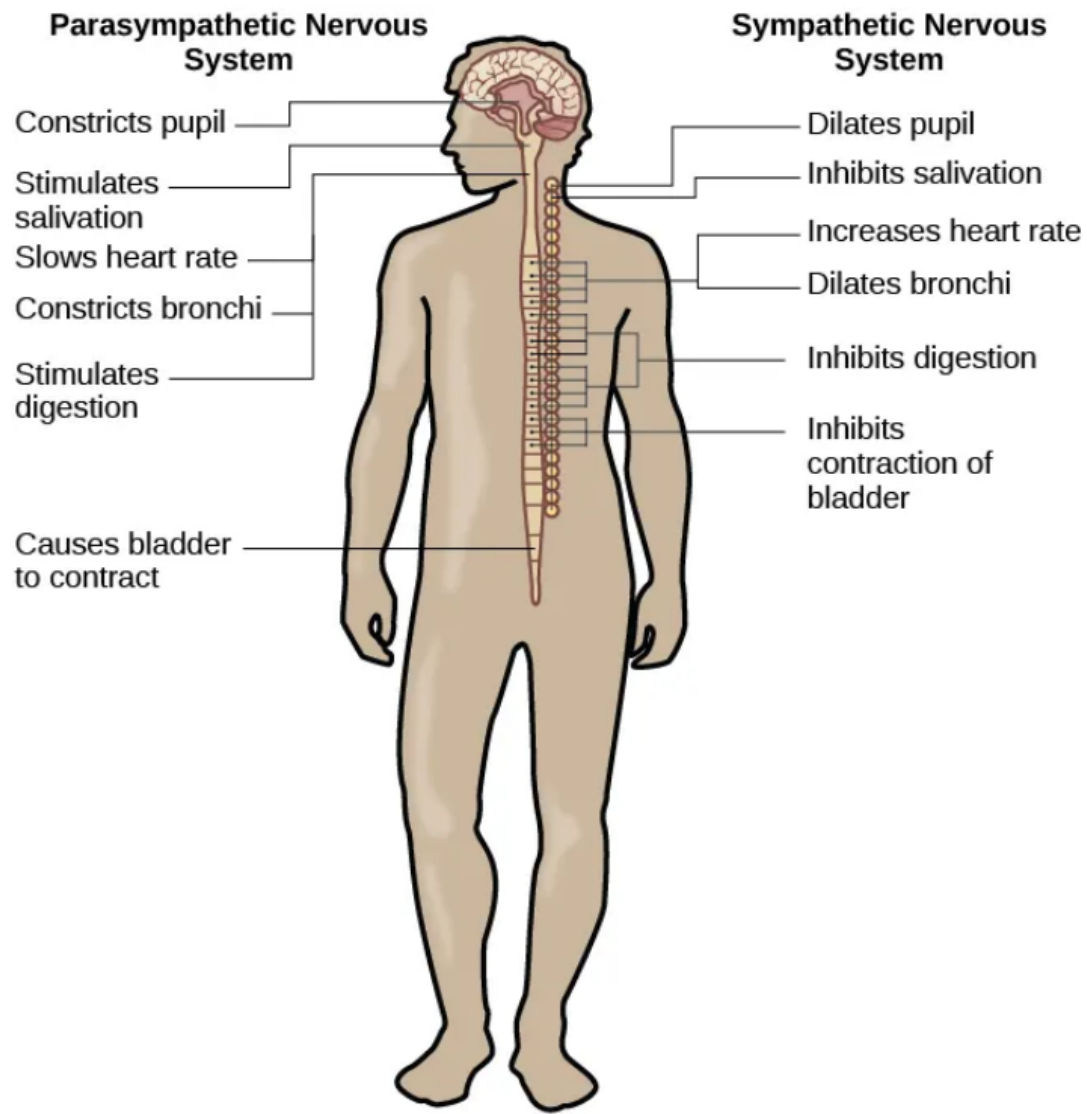
### 4.1 Introduction

- The human nervous system is described between two \_\_\_\_\_ systems:
  - The **central nervous system (CNS)** housing [The Brain and Spinal Cord](#)

- The [Peripheral Nervous System \(PNS\)](#), which is basically everything else in the body that connects back to the CNS

## 4.2 Peripheral Nervous System (PNS)

- The peripheral nervous system has \_\_\_\_\_ of axons called **nerves** which ensure signals from our body are carried to the spine and brain for messaging
- Within the PNS there are two further \_\_\_\_\_:
  - The **somatic nervous system**, which deals with \_\_\_\_\_ or voluntary actions, such as most movements
  - The **autonomic nervous system**, which is related to \_\_\_\_\_ and automatic processes like organ control.
- The somatic nervous system will contain \_\_\_\_\_ neurons that are exiting from the CNS to deliver messages to PNS, to do things like move muscles.
  - On the other hand, there are \_\_\_\_\_ neurons that carry sensory signals back into the CNS to be received in the brain and spinal cord
- The autonomic nervous system has two \_\_\_\_\_ in it as well, the **parasympathetic** and **sympathetic nervous systems**



? Which action would be most associated with the sympathetic nervous system?

- ☒ A) Talking to a friend casually
- ☐ B) Your stomach processing food
- ☐ C) Your hand sending a signal of pain due to being put on a stove
- ☐ D) Running from a lion

Explanation:

## 5 The Brain and Spinal Cord

### 5.1 Introduction

- Making up the CNS, the brain and spinal cord serve essential, core functions in the \_\_\_\_\_ and regulation of our body.
  - The brain can be understood as \_\_\_\_\_ the many different signals, messages, and commands that the body must follow - and has a necessarily complex layout to help accomplish this!

### 5.2 The Spinal Cord

- The spinal cord is the \_\_\_\_\_ and relay station that receives signals from the PNS, but also transmit signals from the brain to the PNS
- It also has it's own built-in system of **reflexes**, which can act without input from the brain

 Discuss: What is an example of a reflex you can think of?

- The spine is \_\_\_\_\_ by vertebrae (bones) and has two types of nerve bundles exiting from it, **sensory** nerves which are ready to receive information and **motor** nerves which connect to muscles and send signals to move

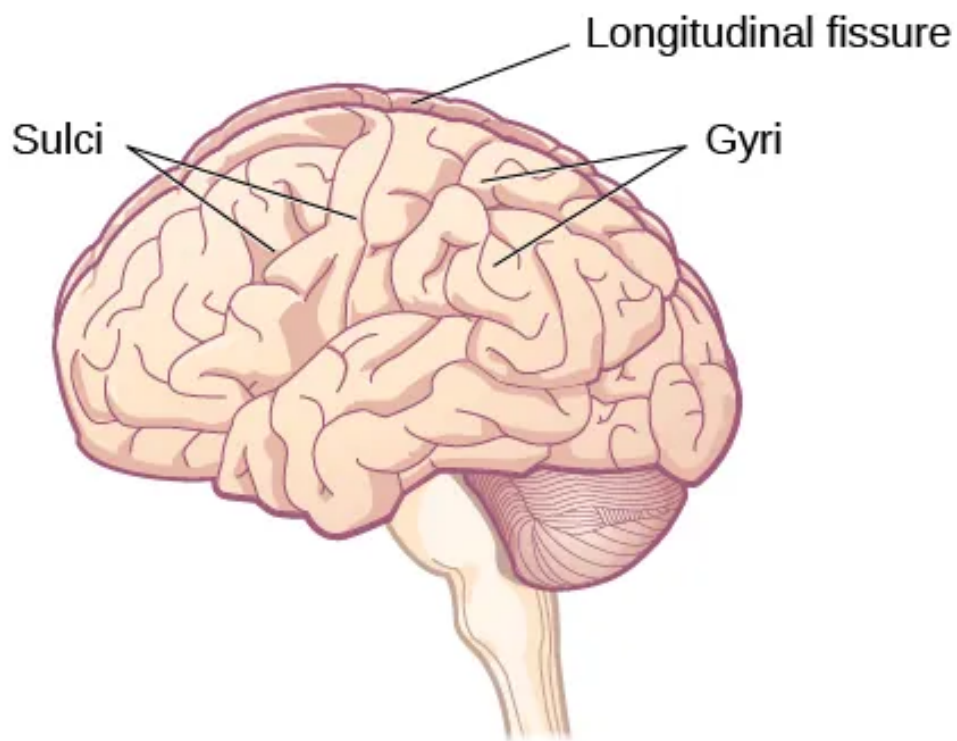
### 5.3 Neuroplasticity

- Injury or problems in development of the CNS can lead to extensive \_\_\_\_\_ in cognition and functioning
  - However, our CNS (and body in general) has a remarkable ability to \_\_\_\_\_ and adapt to damage - this is called **neuroplasticity**
- As a rule of thumb, our brains are able to \_\_\_\_\_ the best when we are young, as our brains are still developing.
  - As we \_\_\_\_\_, our neuroplasticity decreases, but some repair is still possible.

- Neuroplasticity is why some folks can recover a surprising amount of skills after brain \_\_\_\_\_

## 5.4 The Two Hemispheres

- The outside surface of the brain is the **cerebral cortex**, which many uneven grooves and ridges, which allow are cortex to have a large \_\_\_\_\_ area.
  - The bumps are called **gyri** and the \_\_\_\_\_ in between **sulci**, with the most prominent sulcus being the longitudinal fissure that bisect the brain into a left and right half, or **hemisphere**



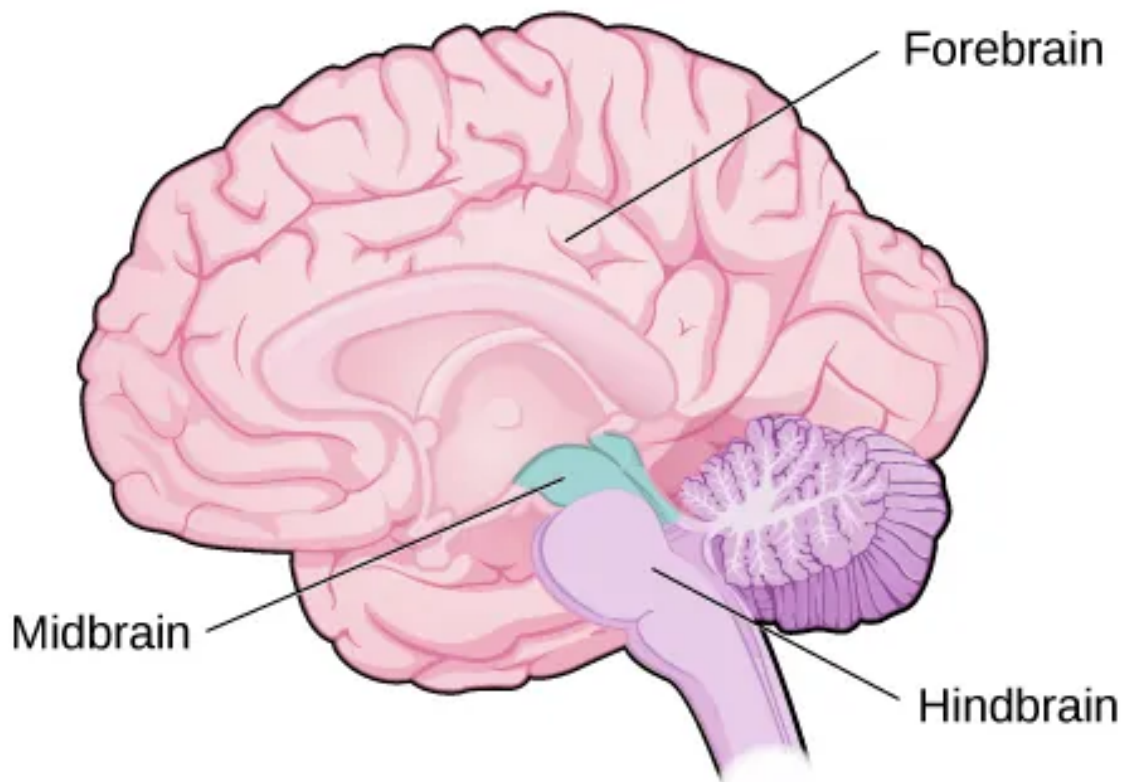
- There is reasonable evidence that certain skills and functions are somewhat \_\_\_\_\_ to a specific region or hemisphere of the brain
  - As a somewhat reductionist example: the left hemisphere generally is more associated with \_\_\_\_\_ emotions and the right hemisphere, more associated with \_\_\_\_\_ emotions

### ! Important

One should be cautious in definitively saying that certain functions only ever exist on one side of the brain - especially on an individual basis



- The \_\_\_\_\_ are connected via the neuron bundle of the **corpus callosum**, which allow communication between the otherwise separate halves of our brain
  - In some cases, this connection may be \_\_\_\_\_ resected or may not develop at all!
  - If the corpus callosum does not form, it results in “split-brain”-ness, which can manifest as poor coordination between separate cognitive skills that are partially lateralized.
- Ironically, most research on the brain's more \_\_\_\_\_ functions is done on individuals that have suffered damage to that area of the brain, and show symptoms that suggest what cognitive abilities may be related to the damaged area



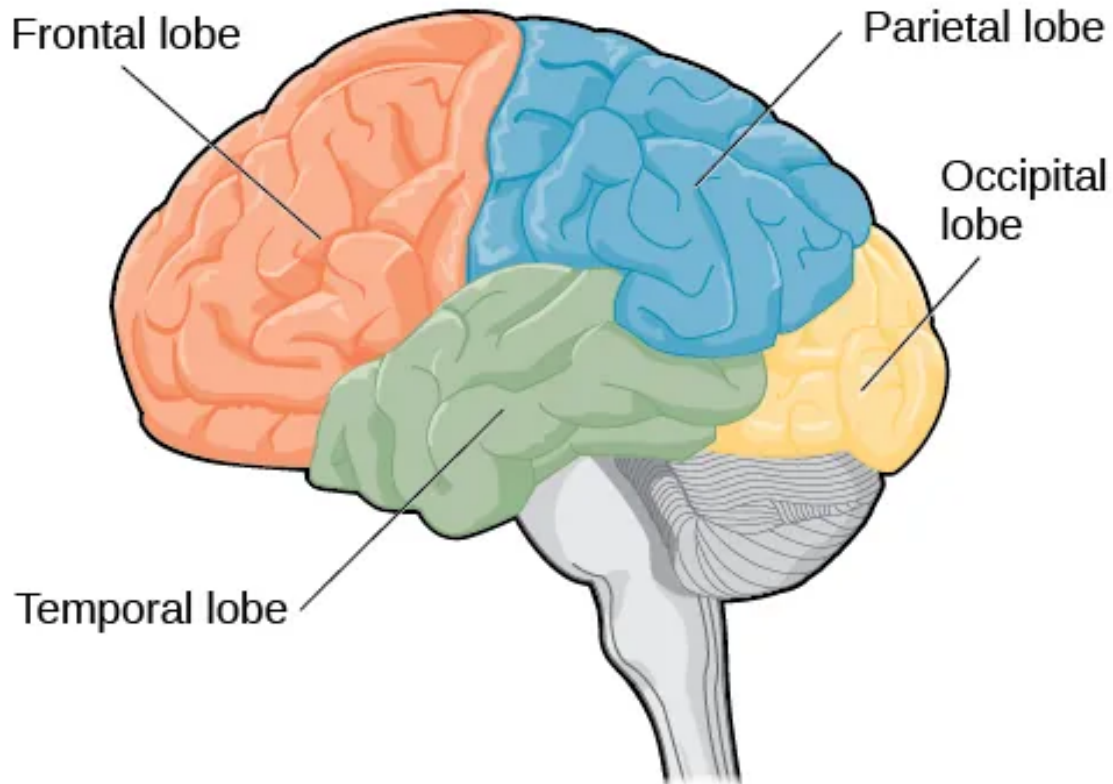
## 5.5 Forebrain Structures

- The **forebrain** is the physically \_\_\_\_\_ part of the brain, visible on the top, front, and most of the side of the brain, and covers the more inner portions
- The following sections will cover the individual components of the forebrain



## 5.6 Lobes of the Brain

- The outermost layer of the forebrain, with the \_\_\_\_\_ and sulci, is called the **cerebral cortex**, and can be separated into 4 distinct **lobes**:
  - The **frontal lobe**
  - The **parietal lobe**
  - The **occipital lobe**
  - The **temporal lobe**



- The frontal lobe is in the front of the brain, behind the \_\_\_\_\_.
  - It is generally involved in “higher-order” processing, \_\_\_\_\_ movement, language, and emotion
  - There are three particularly important areas usually in this area:
    - \* **The motor cortex** which communicates signals to \_\_\_\_\_
    - \* The **prefrontal cortex** which is involved in \_\_\_\_\_ and decision-making
    - \* **Broca’s area**, which is necessary for the \_\_\_\_\_ of speech

? A person suffers damage to their Broca's area in the frontal lobe. What activity are they likely to find difficult?

- ☒ A) Quickly verbally naming off animals
- ☐ B) Pointing to dots on a page
- ☐ C) Recreating a drawing
- ☐ D) Listening to and following complex instructions

Explanation:

- Damage to these area (or the frontal cortex as a whole), is likely to disrupt these functions (see Phineas Gage)



(a)



(b)

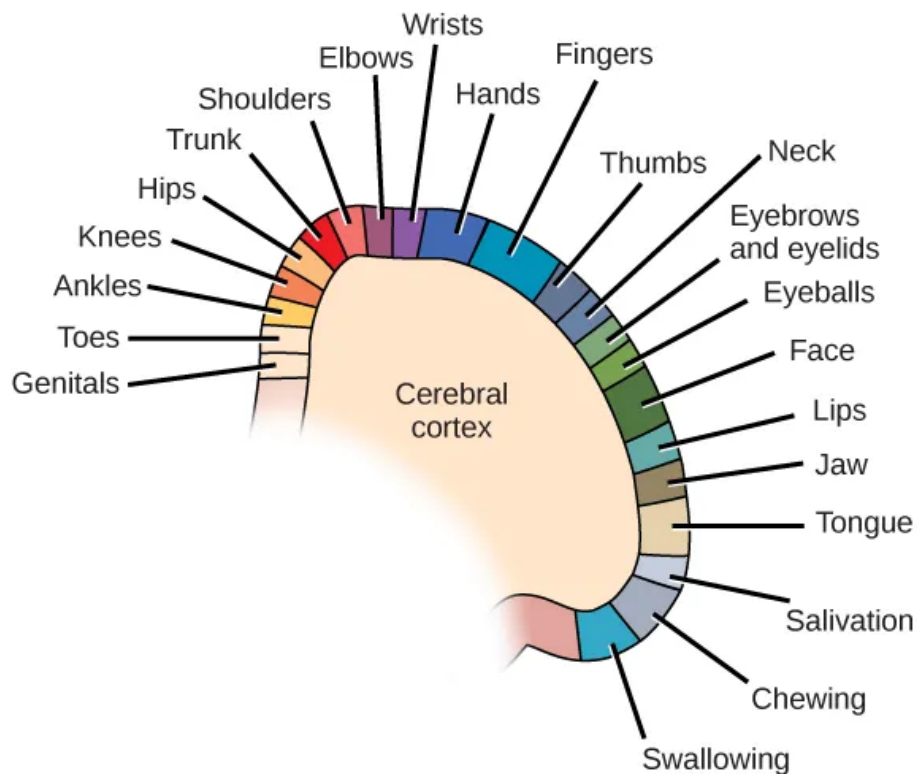
- The **parietal lobe** is primarily used for \_\_\_\_\_ of sensory information via the **somatosensory cortex** - info such as temperature, pain, pressure, etc. is

understood here

? Review: If we are talking about integrating separate sensory information into a single comprehensive perceptual experience, what historical orientation would be most interested in this area?

- ☒ A) Structuralism
- ☐ B) Functionalism
- ☐ C) Behaviorism
- ☐ D) Gestalt

Explanation:



- The **temporal lobe** is to each side of the head, and closely related to \_\_\_\_\_ and memory.
  - It contains the **auditory corex** which processes incoming auditory information, and the **Wernicke's area**, which deals in speech \_\_\_\_\_.
- Finally, **occipital lobe** is at the back of the brain and is responsible for integrating \_\_\_\_\_ information

## 5.7 Other Areas of the Forebrain

- Underneath the \_\_\_\_\_ are the **subcortical structures** that also serve important roles in cognition
- The **thalamus** handle \_\_\_\_\_ information about senses (except smell) to the correct areas of the cortex for further processing
- The **limbic system** is made up of several different components, all related to \_\_\_\_\_ and memory.
  - The **hippocampus** is used for learning and memory
  - The **amygdala** is necessary for \_\_\_\_\_ memory and emotions
  - The **hypothalamus** helps regulate homeostasis, especially via sending signals to [The Endocrine System](#).

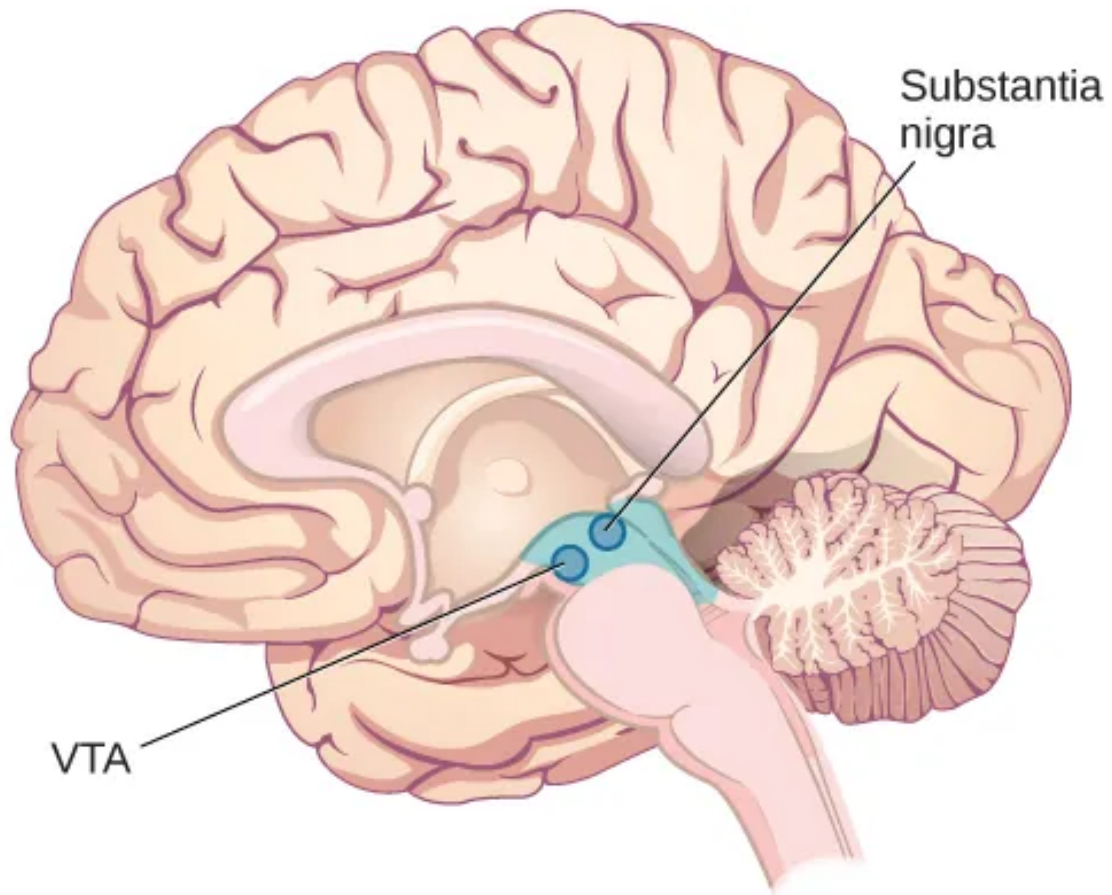
? A child runs away from their parent across a busy street, a very poor and dangerous decision - what are of the brain is likely undeveloped in the child?

- ☒ A) Thalamus
- ☐ B) Hypothalamus
- ☐ C) Prefrontal Cortex
- ☐ D) Somatosensory Cortex

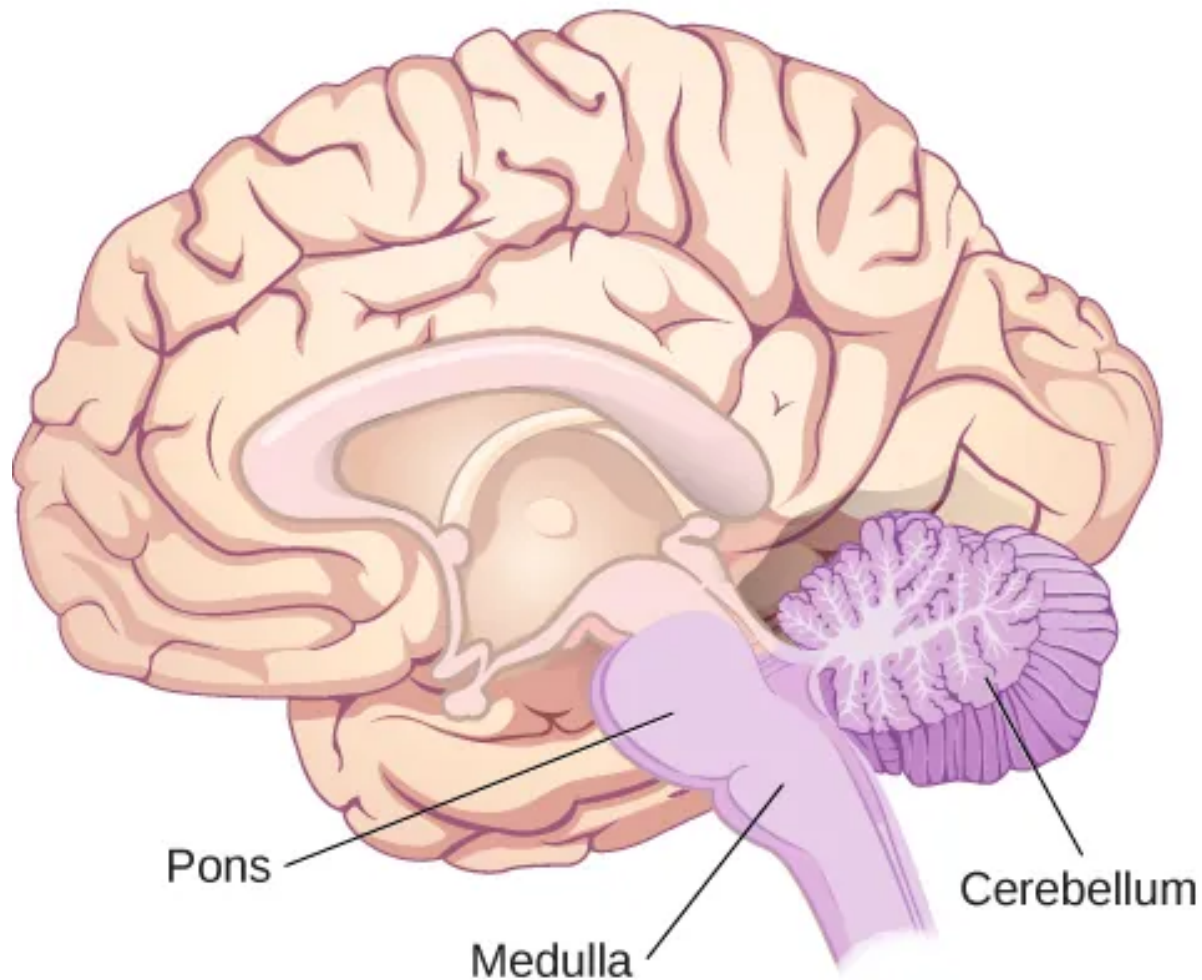
Explanation:

## 5.8 Midbrain and Hindbrain Structures

- The **midbrain** rests \_\_\_\_\_ the forebrain, and in front of the **hindbrain**.
  - The **reticular formation** extend all the way to the edge of the forebrain, and goes down to the start of the hindbrain - it plays an important role in \_\_\_\_\_, sleep, and arousal
  - The **substantia nigra** and **ventral tegmental area (VTA)** indirectly help ensure movement goes \_\_\_\_\_, by producing the dopamine neurotransmitter
    - \* These structures become degraded in Parkinson's disease, contributing to the \_\_\_\_\_ problems associated with that disorder



- The **hindbrain** is effectively the connecting point between the spine and brain, and contains several specific structures
  - The **medulla oblongata** (also just called the medulla) handles \_\_\_\_\_ breathing, blood pressure, and heart rate
  - The **Pons** is most involved in regulation of \_\_\_\_\_
  - The **Cerebellum** hangs off the back of the brainstem, and mostly helps coordination, balance, \_\_\_\_\_ - but also some motor-based procedural memory



## 5.9 Brain Imaging Introduction

- Brain imaging has \_\_\_\_\_ greatly in the last several decades; which has allowed researchers to establish a much better understanding of structural \_\_\_\_\_ in the brain while a person is still alive
  - While autopsy was always an option, it was/is \_\_\_\_\_ to see how the brain works when it is non-functioning

## 5.10 Techniques Involving Radiation

- A **Computerized tomography (CT)** scan is effectively a series of \_\_\_\_\_ across the layers of the brain, eventually resulting in a complete image of the brain structures
  - This was one of the \_\_\_\_\_ technologies to scan brains, and continues to see use today despite being less detailed than MRIs.



- A **Positron emission tomography (PET)** uses a \_\_\_\_\_ “tracer” that flows with the blood in the brain to determine how certain areas “activate” i.e., get more blood - it is not used as much on it's own now, given that it can't visualize structure very well, but is sometimes used in conjunction with CTs.

 Discuss: If a CT scan is less detailed than newer methods, why do you think it would be used so commonly?

## 5.11 Techniques Involving Magnetic Fields

- **Magnetic Resonance Imaging (MRI)** uses the existing \_\_\_\_\_ within us and a magnetic field to detect tissue density and visualize structures.
  - In concept, they produce a similar static, structural image of the brain as a \_\_\_\_\_ scan, but are more detailed.
- **Functional magnetic resonance imaging (fMRI)** are even more detailed and show activity over \_\_\_\_\_ in the brain, much like with a PET scan, but with far more structural detail at the same time.
  - Like with PET scans, fMRIs use blood flow to track movement, but also use the MRI \_\_\_\_\_ imaging at the same time

 Discuss: Why don't we just do fMRIs on everyone to check how healthy people's brain are? What barriers exist to that plan?

## 5.12 Techniques Involving Electrical Activity

- **Electroencephalography (EEG)** scans use \_\_\_\_\_ on the skull to track electrical activity in certain areas of the brain

- Used a conductive gel and skull cap, and the individual electrodes can show areas of \_\_\_\_\_ high and low energy
- Especially useful in sleep and \_\_\_\_\_ analysis, as it can be useful to see what areas of the brain are showing an excess or absence of activity

## 6 The Endocrine System

### 6.1 Introduction

- The **endocrine system** is responsible for the \_\_\_\_\_ and regulation of chemical **hormones** throughout the body.
  - Hormones do not travel via neuron action \_\_\_\_\_, but instead travel in the bloodstream, having systemic effects across the body
  - The **hypothalamus** (in the CNS) and the \_\_\_\_\_ gland (of the endocrine system) work together to ensure hormones are in the correct concentration - as imbalance can be harmful/dangerous

### 6.2 Major Glands

#### ! Important

Dysfunction in any of these glands can cause major disruption to mood, fatigue, and other mental functions

- The **pituitary gland** serves as the \_\_\_\_\_ to all the hormone creation occurring in the other endocrine system glands, working on instructions from the hypothalamus
  - It \_\_\_\_\_ how the instructions are sent and carried to the other glands
- The **thyroid gland** plays an important role in \_\_\_\_\_ growth, metabolism, and appetite
- The **adrenal glands** \_\_\_\_\_ produces epinephrine and norepinephrine, to control our adrenaline levels
- The **pancreas** is responsible for regulation of blood \_\_\_\_\_ and insulin throughout the body
  - Poor regulation in the pancreas or it's products results in **diabetes**, meaning the additional \_\_\_\_\_ must be taken by the person or medicine to maintain homeostasis.



- The **gonads** are sex-specific, and create hormones appropriate to the sexual function of the \_\_\_\_\_ that are produced in one's body

Major Endocrine Glands and Associated Hormone Functions

| Endocrine Gland | Associated Hormones                                                                     | Function                                              |
|-----------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------|
| Pituitary       | Growth hormone, releasing and inhibiting hormones (such as thyroid stimulating hormone) | Regulate growth, regulate hormone release             |
| Thyroid         | Thyroxine, triiodothyronine                                                             | Regulate metabolism and appetite                      |
| Pineal          | Melatonin                                                                               | Regulate some biological rhythms such as sleep cycles |
| Adrenal         | Epinephrine, norepinephrine                                                             | Stress response, increase metabolic activities        |
| Pancreas        | Insulin, glucagon                                                                       | Regulate blood sugar levels                           |
| Ovaries         | Estrogen, progesterone                                                                  | Mediate sexual motivation and behavior, reproduction  |
| Testes          | Androgens, such as testosterone                                                         | Mediate sexual motivation and behavior, reproduction  |

? Using the table and information above, what is the pathway for messaging the release of glucagon?

- ☒ A) Pituitary -> Thyroid -> Pineal  
☐ B) Pituitary -> Thyroid -> Pancreas  
☐ C) Hypothalamus -> Pituitary -> Testes  
☐ D) Hypothalamus -> Pituitary -> Pancreas

Explanation:

## 7 Conclusion

### 7.1 Recap

- While it may feel somewhat removed from what we talk about in future modules, it is important to understand that our genetics, anatomy, and physiology play a crucial role in our behaviors and cognition
- The Biopsychology perspective is one that is often integrated closely with neuroscience, chemistry, and biology; as you take classes in these other domains, you'll likely see overlap between what we talked about and those other disciplines
- Try not to get too weighed down in the biological and chemical terminology when studying this module - primarily know what sorts of cognition, processing, and behaviors are associated with which structures

### 7.2 Lecture Check-in

- Get into assigned groups for our weekly group work activity!